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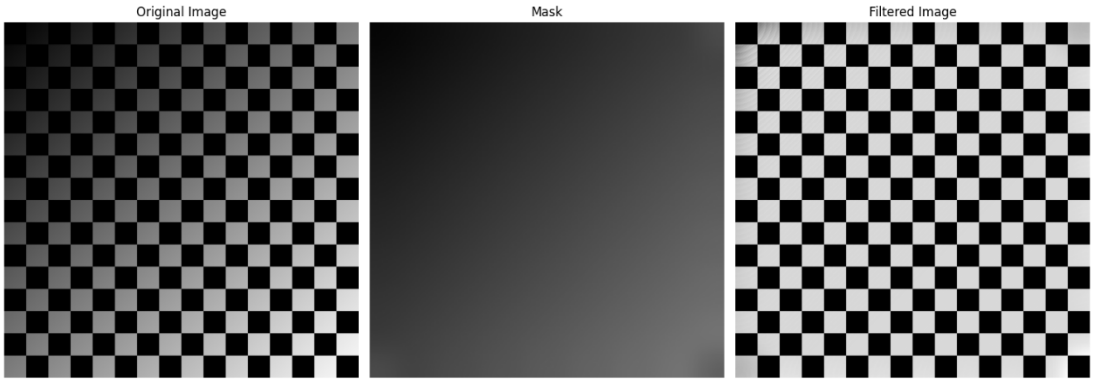
1.

Checkerboard:

Gaussian kernel specs: Width = 256, Height = 256, K = 1, standard deviation = 64

Result:

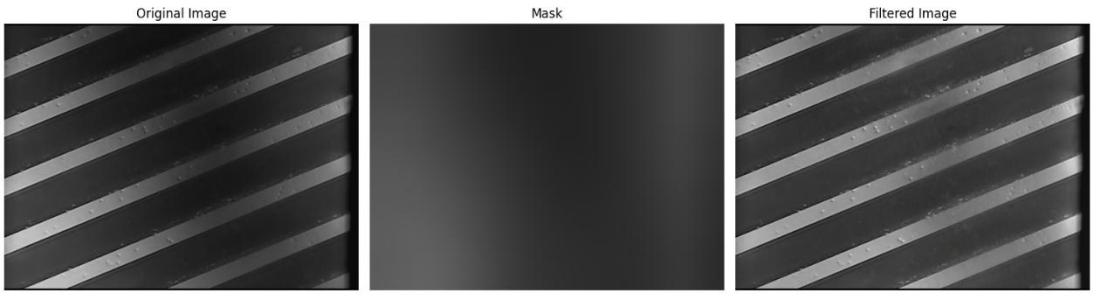


2.

N1:

Gaussian kernel specs: Width = 160, Height = 480, K = 1, standard deviation for width = 40, standard deviation for height = 120

Result:



3.

For the checkerboard image, as the image has a resolution of 1024x1024, & each square is a 64*64 pixel. For this problem, a 256x256 Gaussian low-pass filter kernel with a standard deviation of 64. A circularly symmetric 2d Gaussian is designed as the darker area seems to be circular. As we can see from question 1, using a Gaussian kernel with these specifications helps with removing the darker region of the image.

For the N1 image, as the resolution of the image is 640x480, the darker region of the image seems to be long but skinny, so we use a 160x480 Gaussian low-pass Filter Kernel with a standard deviation of 40 & 120. As we can see from question 2, using a Gaussian kernel with these specifications helps with removing the darker region of the image.

From the design of the Gaussian low pass filter, we must consider the size & shape of what we want to filter by adjusting the width, height & also the standard deviation accordingly.